STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-LIFT-EQUIVALENCE-LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2531-2540 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 2542-2597, the complete author proof (54 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_examples-defos.json (pb=2795, stmt 2531-2540, proof 2542-2597).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00DV: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2597–2597. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZ3: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2293–2300. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZ2: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2213–2234. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07M8: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2597–2597. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A' \to A$ be a surjective map of Noetherian rings with nilpotent kernel. Let $A \to B$ be a finite type flat ring map. Let $S \subset B$ be a multiplicative subset such that if $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is not syntomic at $\mathfrak{q}$, then $S \cap \mathfrak{q} = \emptyset$. Then B has a flat lift to A' if and only if $S^{-1}B$ has a flat lift to A'.*

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This proof is the same as the proof of Lemma 0DZ3 but easier. We suggest the reader to skip the proof. Choose an A-algebra surjection $P = A[x_1, \ldots, x_n] \to B$. Let $S_P \subset P$ be the inverse image of S. Set $P' = A'[x_1, \ldots, x_n]$ and denote $S_{P'} \subset P'$ the inverse image of S_P.

Suppose $A' \to B'$ is a flat lift of $A \to B$, in other words, $A' \to B'$ is flat and there is an A-algebra isomorphism $B = B' \otimes_{A'} A$. Then we can choose an A'-algebra map $P' \to B'$ lifting the given surjection $P \to B$. By Nakayama's lemma (Algebra, Lemma 00DV) we find that B' is a quotient of P'. In particular, we find that we can endow B' with an A'-flat P'-module structure lifting B as an A-flat P-module. Conversely, if we can lift B to a P'-module M' flat over A', then M' is a cyclic module $M' \cong P'/J'$ (using Nakayama again) and setting $B' = P'/J'$ we find a flat lift of B as an algebra.

Set $C = S^{-1}B$. Suppose that $A' \to C'$ is a flat lift of $A \to C$. Elements of C' which map to invertible elements of C are invertible. We choose an A'-algebra map $P' \to C'$ lifting the A-algebra map $P \to C$. By the remark above these maps pass through localizations to give surjections $S_P^{-1}P \to C$ and $S_{P'}^{-1}P' \to C'$ (for the second use Nakayama's lemma). In particular, we find that we can endow C' with an A'-flat $S_{P'}^{-1}P'$-module structure lifting C as an A-flat $S_P^{-1}P$-module. Conversely, if we can lift C to a $S_{P'}^{-1}P'$-module N' flat over A', then N' is a cyclic module $N' \cong S_{P'}^{-1}P'/\tilde{J}$ (using Nakayama again) and setting $C' = S_{P'}^{-1}P'/\tilde{J}$ we find a flat lift of C as an algebra.

The syntomic locus of a morphism of schemes is open by definition. Let $J_B \subset B$ be an ideal cutting out the set of points in $\mathrm{Spec}(B)$ where $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is not syntomic. Denote $J_P \subset P$ and $J_{P'} \subset P'$ the corresponding ideals. Observe that $P' \to S_{P'}^{-1}P'$ is a flat ring map which induces an isomorphism $P'/J_{P'} = S_{P'}^{-1}P'/J_{P'}S_{P'}^{-1}P'$ by our assumption on S in the lemma, namely, the assumption in the lemma is exactly that $B/J_B = S^{-1}(B/J_B)$. We conclude that our lemma is a consequence of Lemma 0DZ2 provided we can show that B_g lifts to an A'-flat P'_g-module for $g \in J_B$. However, the ring map $A \to B_g$ is syntomic and hence lifts to an A'-flat algebra B' by Smoothing Ring Maps, Proposition 07M8. Since $A' \to P'_g$ is smooth, we can lift $P_g \to B_g$ to a surjective map $P'_g \to B'$ as before and we get what we want. □